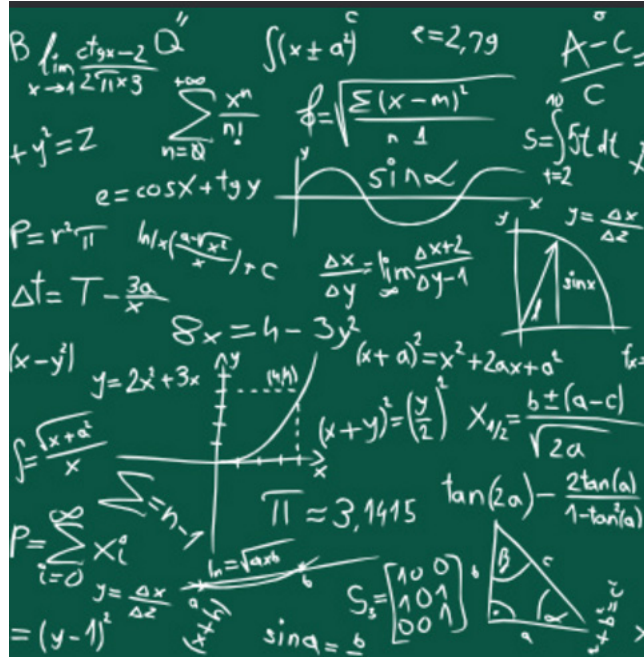


Computing limits of sequences



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- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Computing limits of sequences

- The proofs of the **basic theorems** on computing limits are based on the **definition of limit**, on the **use of inequalities**, and on the use of **properties that are eventually true**.
- In particular, these theorems illustrate the relation between the limit operation and the algebraic structures¹ and the order structures² present in $\mathbb{R}$.

We will simply write $a_n \rightarrow \ell$ to mean $a_n \rightarrow \ell$ for $n \rightarrow +\infty$

Properties of the limit operation with respect to algebraic operations.

Theorem 1: Algebra of limits, case of finite limits

Hypotheses:

$$1. a_n \rightarrow \ell_a \in \mathbb{R} \quad 2. b_n \rightarrow \ell_b \in \mathbb{R}.$$

Claim:

$$\begin{aligned} 1. a_n \pm b_n &\rightarrow \ell_a \pm \ell_b & 2. \frac{a_n}{b_n} &\rightarrow \frac{\ell_a}{\ell_b} \quad (b_n, \ell_b \neq 0, \text{ eventually}) \\ 3. a_n b_n &\rightarrow \ell_a \ell_b & 4. a_n^{b_n} &\rightarrow \ell_a^{\ell_b} \quad (a_n, \ell_a > 0, \text{ eventually}). \end{aligned}$$

Proof. We prove that:

$$a_n \rightarrow \ell_a \in \mathbb{R}, b_n \rightarrow \ell_b \in \mathbb{R} \Rightarrow a_n + b_n \rightarrow \ell_a + \ell_b$$

We consider

$$|(a_n + b_n) - (\ell_a + \ell_b)| = |(a_n - \ell_a) + (b_n - \ell_b)| \leq |a_n - \ell_a| + |b_n - \ell_b| \quad (1)$$

by the triangle inequality. Since by hypothesis $a_n \rightarrow \ell_a$ and $b_n \rightarrow \ell_b$, we have that

$$|a_n - \ell_a| < \varepsilon_a \quad \text{and} \quad |b_n - \ell_b| < \varepsilon_b \quad (\text{eventually})$$

for every $\varepsilon_a > 0$ and $\varepsilon_b > 0$. Bounding from above the terms on the right-hand side of (1) we conclude that

$$|(a_n + b_n) - (\ell_a + \ell_b)| < \underbrace{2 \max\{\varepsilon_a, \varepsilon_b\}}_{=\tilde{\varepsilon} \text{ and } >0}$$

Since $\tilde{\varepsilon}$ is arbitrary, the claim follows. □

¹A set on which an operation is defined is called an algebraic structure

²A set is endowed with an order structure if an order relation is defined on it

Proof. We prove that:

$$a_n \rightarrow \ell_a \in \mathbb{R}, b_n \rightarrow \ell_b \in \mathbb{R} \Rightarrow a_n - b_n \rightarrow \ell_a - \ell_b$$

For every $\varepsilon_a > 0$ and $\varepsilon_b > 0$, we have:

$$|(a_n - b_n) - (\ell_a - \ell_b)| = |(a_n - \ell_a) + (\ell_b - b_n)| \leq \underbrace{|a_n - \ell_a|}_{< \varepsilon_a} + \underbrace{|\ell_b - b_n|}_{= |b_n - \ell_b| < \varepsilon_b}$$

Hence

$$|(a_n - b_n) - (\ell_a - \ell_b)| < \underbrace{2 \max\{\varepsilon_a, \varepsilon_b\}}_{=\tilde{\varepsilon} \text{ and } >0}$$

Since $\tilde{\varepsilon}$ is arbitrary, the claim follows. □

Proof. We prove that:

$$a_n \rightarrow \ell_a \in \mathbb{R}, b_n \rightarrow \ell_b \in \mathbb{R} \Rightarrow a_n b_n \rightarrow \ell_a \ell_b$$

We consider

$$\begin{aligned} |(a_n b_n) - (\ell_a \ell_b)| &= |a_n b_n - a_n \ell_b + a_n \ell_b - \ell_a \ell_b| \\ &= |a_n(b_n - \ell_b) + \ell_b(a_n - \ell_a)| \leq |a_n(b_n - \ell_b)| + |\ell_b(a_n - \ell_a)| \\ &= |a_n| |b_n - \ell_b| + |\ell_b| |a_n - \ell_a| \end{aligned}$$

by the triangle inequality and the properties of the absolute value. Hence

$$|(a_n b_n) - (\ell_a \ell_b)| \leq |a_n| |b_n - \ell_b| + |\ell_b| |a_n - \ell_a| \quad (2)$$

Since by hypothesis $a_n \rightarrow \ell_a$ and $b_n \rightarrow \ell_b$, for every $\varepsilon_a > 0$ and $\varepsilon_b > 0$ we have

$$|a_n - \ell_a| < \varepsilon_a \quad \text{and} \quad |b_n - \ell_b| < \varepsilon_b \quad (\text{eventually}).$$

Moreover, since

$$|a_n - \ell_a| \geq |a_n| - |\ell_a| \quad \text{we have} \quad |a_n| < |\ell_a| + \varepsilon_a, \quad \text{eventually.}$$

Therefore, bounding from above the terms on the right-hand side of (2), we conclude that

$$|(a_n b_n) - (\ell_a \ell_b)| < (|\ell_a| + \varepsilon_a) \varepsilon_b + |\ell_b| \varepsilon_a = \underbrace{|\ell_a| \varepsilon_b + |\ell_b| \varepsilon_a + \varepsilon_a \cdot \varepsilon_b}_{=\tilde{\varepsilon} \text{ and } >0}$$

Since $\tilde{\varepsilon}$ is arbitrary, the claim follows. □

- Moreover, the **limit operation preserves the ordering**

Theorem 2: Sign-preservation, first form

Hypotheses:

$$1. a_n \rightarrow \ell_a \quad 2. \ell_a \leq 0.$$

Claim:

$$a_n \leq 0, \text{ eventually.}$$

Proof. We consider the case $\ell_a > 0$. By definition of limit we have that

$$|a_n - \ell_a| < \varepsilon, \quad \text{eventually,}$$

for every $\varepsilon > 0$, which we rewrite in the form:

$$\ell_a - \varepsilon < a_n < \ell_a + \varepsilon, \quad \text{eventually.}$$

Since $\ell_a > 0$, we can choose $\varepsilon > 0$ such that $\ell_a - \varepsilon > 0$; then the inequality

$$0 < \ell_a - \varepsilon < a_n$$

shows that $a_n > 0$, eventually.

The case $\ell_a < 0$ is proved analogously. □

Theorem 3: Sign-preservation, second form (part I)

Hypotheses:

$$1. \ a_n \rightarrow \ell_a \in \mathbb{R} \quad 2. \ a_n \geq 0, \text{ eventually.}$$

Claim:

$$\ell_a \geq 0.$$

Proof. It follows from the previous theorem. Indeed, if by contradiction we had $\ell_a < 0$, from the previous theorem we would have $a_n < 0$ eventually, which is incompatible with the hypothesis that $a_n \geq 0$ eventually.

This case cannot occur, i.e., the opposite holds, which is the claim of the theorem. □

Theorem 4: Sign-preservation, second form (part II)

Hypotheses:

$$1. \ a_n \rightarrow \ell_a \in \mathbb{R} \quad 2. \ b_n \rightarrow \ell_b \in \mathbb{R} \quad 3. \ a_n \geq b_n, \text{ eventually.}$$

Claim:

$$\ell_a \geq \ell_b.$$

Proof. We consider the sequence $a_n - b_n$; by the theorem on the algebra of limits we have

$$a_n - b_n \rightarrow \ell_a - \ell_b.$$

Since by hypothesis

$$a_n \geq b_n \text{ and hence } a_n - b_n \geq 0,$$

by the sign-preservation theorem, second form (part I), applied to the sequence $a_n - b_n$ we have

$$\ell_a - \ell_b \geq 0 \text{ and hence } \ell_a \geq \ell_b.$$

□

- This theorem tells us that in an inequality between two sequences we can pass to the limit on both sides, keeping the “ $\leq$ ” or the “ $\geq$ ”.
- Note, instead, that in general strict inequalities “ $<$ ” and “ $>$ ” are not preserved when passing to the limit.

Example 1: Passing to the limit with strict inequalities

For example, even if the a_n are strictly positive, their limit ℓ_a is positive or zero, as shown by the simple example $\frac{1}{n} \rightarrow 0$.

Theorem 5: Comparison (squeeze) theorem

Hypotheses:

$$1. \ a_n \rightarrow \ell, c_n \rightarrow \ell \text{ and } \ell \in \mathbb{R}, \quad 2. \ a_n \leq b_n \leq c_n, \text{ eventually.}$$

Claim:

$$b_n \rightarrow \ell.$$

Proof. By definition of limit we have, eventually, that

$$\ell - \varepsilon_a < a_n < \ell + \varepsilon_a \quad \text{and} \quad \ell - \varepsilon_c < c_n < \ell + \varepsilon_c$$

for every $\varepsilon_a > 0$ and $\varepsilon_c > 0$. Hence:

$$\ell - \max\{\varepsilon_a, \varepsilon_c\} < a_n < \ell + \max\{\varepsilon_a, \varepsilon_c\} \quad \text{and} \quad \ell - \max\{\varepsilon_a, \varepsilon_c\} < c_n < \ell + \max\{\varepsilon_a, \varepsilon_c\}$$

From the hypotheses of the theorem we therefore have, eventually, that

$$\ell - \max\{\varepsilon_a, \varepsilon_c\} < a_n \leq b_n \leq c_n < \ell + \max\{\varepsilon_a, \varepsilon_c\}$$

But then, eventually, we have

$$\ell - \underbrace{\max\{\varepsilon_a, \varepsilon_c\}}_{=\tilde{\varepsilon} \text{ and } >0} < b_n < \ell + \underbrace{\max\{\varepsilon_a, \varepsilon_c\}}_{=\tilde{\varepsilon} \text{ and } >0}.$$

Since $\tilde{\varepsilon}$ is arbitrary, the claim follows. □

- Frequently used special cases of this theorem are expressed by the following corollaries, which are very useful when studying the product of an oscillating (but bounded) sequence and one that tends to zero

Corollary 1: Of the comparison theorem (part I)

Hypotheses:

1. $c_n \rightarrow 0$
2. $|b_n| \leq c_n$, eventually.

Claim:

$$b_n \rightarrow 0.$$

Proof. We know that eventually we have $-c_n \leq b_n \leq c_n$. Obviously

$$\text{if } c_n \rightarrow 0 \text{ then } -c_n \rightarrow 0.$$

Hence by the comparison theorem (with $a_n = -c_n$ and $\ell = 0$) we have that $b_n \rightarrow 0$. □

Corollary 2: Of the comparison theorem (part II)

Hypotheses:

1. $c_n \rightarrow 0$ 2. $\{b_n\}$ is bounded (but not necessarily convergent).

Claim:

$$c_n b_n \rightarrow 0.$$

Proof. If $\{b_n\}$ is bounded, then $|b_n| \leq M$ for some $M > 0$ and for every $n \in \mathbb{N}$. We can therefore write

$$|b_n c_n| \leq M |c_n|.$$

Since

$$c_n \rightarrow 0 \text{ also } M |c_n| \rightarrow 0,$$

by Corollary (??) we conclude that $b_n c_n \rightarrow 0$. □

The product of an infinitesimal sequence and a bounded one is infinitesimal.

Example 2: Application of the corollary

Consider the sequence given by a ratio of two expressions, each consisting of a sum of powers of n , such as:

$$n \mapsto \frac{n^{5/2} - 3n + 7}{n^3 + \sqrt{n} - 3n^2}$$

Factoring out the highest power both in the numerator and in the denominator we obtain:

$$\frac{n^{5/2} \left(1 - \frac{3}{n^{3/2}} + \frac{7}{n^{5/2}}\right)}{n^3 \left(1 + \frac{1}{n^{5/2}} - \frac{3}{n}\right)} = \frac{1}{\sqrt{n}} \frac{1 - \frac{3}{n^{3/2}} + \frac{7}{n^{5/2}}}{1 + \frac{1}{n^{5/2}} - \frac{3}{n}}$$

Now, by Theorem (??) on the algebra of limits and knowing that negative powers of n tend to zero, we can state that:

$$1 - \underbrace{\frac{3}{n^{3/2}}}_{\rightarrow 0} + \underbrace{\frac{7}{n^{5/2}}}_{\rightarrow 0} \rightarrow 1, \quad 1 + \underbrace{\frac{1}{n^{5/2}}}_{\rightarrow 0} - \underbrace{\frac{3}{n}}_{\rightarrow 0} \rightarrow 1 \quad \text{and} \quad \left(\frac{1 - \frac{3}{n^{3/2}} + \frac{7}{n^{5/2}}}{1 + \frac{1}{n^{5/2}} - \frac{3}{n}} \right) \rightarrow 1$$

hence the last sequence is convergent and consequently bounded. Now, by Corollary (??) and since

$$\frac{1}{\sqrt{n}} \rightarrow 0 \text{ we have } \frac{n^{5/2} - 3n + 7}{n^3 + \sqrt{n} - 3n^2} \rightarrow 0$$

Example 3: Application of the corollary

The sequence

$$n \mapsto \frac{\sin n}{n}$$

is the product of two sequences

$$n \mapsto \frac{1}{n} \text{ (convergent, infinitesimal) and } n \mapsto \sin n \text{ (irregular),}$$

hence Theorem (??) on the algebra of limits cannot be applied (the second limit does not exist).

- However, Corollary (??) can be applied. The sequence $\{\frac{1}{n}\}$ is infinitesimal and, since $|\sin n| \leq 1$, the sequence $\{\sin n\}$ is bounded; therefore we have

$$\lim_{n \rightarrow +\infty} \frac{\sin n}{n} = 0$$

- So far we have seen theorems that work on pairs of sequences that are both convergent or at least bounded.

Sequences with limits $+\infty$ and $-\infty$

- Suppose, for example, that

$$a_n \rightarrow \ell_a \text{ and } b_n \rightarrow +\infty$$

then it is easy (and intuitive) to see that

$$a_n + b_n \rightarrow +\infty$$

We will abbreviate this as follows:

$$\ell_a + \infty = +\infty$$

- Reasoning in a similar way we can summarize the rules for the limit of the sum (or difference) of two sequences, one or both of which are divergent.

Rules of partial arithmetization of the infinity symbol

Theorem 6: Partial arithmetization of the infinity symbol (addition)

Hypotheses:

$$1. a_n \rightarrow \ell_a \in \mathbb{R} \quad 2. b_n \rightarrow +\infty \quad 3. c_n \rightarrow +\infty.$$

Claim:

$$\begin{aligned} 1. a_n + b_n &\rightarrow \ell_a + \infty = +\infty & 2. a_n - b_n &\rightarrow \ell_a - \infty = -\infty \\ 3. b_n + c_n &\rightarrow +\infty + \infty = +\infty & 4. -b_n - c_n &\rightarrow -\infty - \infty = -\infty. \end{aligned}$$

Theorem 7: Partial arithmetization of the infinity symbol (product)

Hypotheses:

$$1. a_n \rightarrow \ell_a \in \mathbb{R} \quad 2. b_n \rightarrow 0 \quad 3. c_n \rightarrow \infty.$$

Claim:

$$1. a_n c_n \rightarrow \ell_a \infty = \infty \quad (\ell_a \neq 0) \quad 2. \frac{a_n}{b_n} \rightarrow \frac{\ell_a}{0} = \infty \quad (\ell_a \neq 0) \quad 3. \frac{a_n}{c_n} \rightarrow \frac{\ell_a}{\infty} = 0.$$

- the **sign** of ∞ must be determined with the **usual rule of signs**.

Example 4: Rule of signs

We have:

- if $a_n \rightarrow \ell_a \in \mathbb{R}, \ell_a > 0$ and $b_n \rightarrow 0^+$ then $\frac{a_n}{b_n} \rightarrow +\infty$
- if $a_n \rightarrow \ell_a \in \mathbb{R}, \ell_a < 0$ and $b_n \rightarrow 0^-$ then $\frac{a_n}{b_n} \rightarrow +\infty$
- if $a_n \rightarrow \ell_a \in \mathbb{R}, \ell_a > 0$ and $b_n \rightarrow 0^-$ then $\frac{a_n}{b_n} \rightarrow -\infty$
- if $a_n \rightarrow \ell_a \in \mathbb{R}, \ell_a < 0$ and $b_n \rightarrow 0^+$ then $\frac{a_n}{b_n} \rightarrow -\infty$

Therefore, in order to apply the rules of partial arithmetization of the infinity symbol, it is necessary to determine whether b_n tends to zero from above or from below.

Proof. We prove that:

$$a_n \rightarrow \ell_a \in \mathbb{R}, c_n \rightarrow +\infty \Rightarrow \frac{a_n}{c_n} \rightarrow 0.$$

For every $\varepsilon > 0$, since $a_n \rightarrow \ell_a$, eventually we have

$$\underbrace{|a_n - \ell_a|}_{\geq |a_n| - |\ell_a|} < \varepsilon \quad \text{hence} \quad |a_n| < |\ell_a| + \varepsilon.$$

Moreover, since $c_n \rightarrow +\infty$, eventually we have

$$c_n > \frac{1}{\varepsilon}$$

It follows that, eventually, we have

$$\frac{|a_n|}{|c_n|} < \frac{|\ell_a| + \varepsilon}{\frac{1}{\varepsilon}} \quad \text{and hence} \quad \left| \frac{a_n}{c_n} \right| < \varepsilon (|\ell_a| + \varepsilon) = \underbrace{|\ell_a| \varepsilon + \varepsilon^2}_{=\tilde{\varepsilon} \text{ and } > 0}$$

Since $\tilde{\varepsilon}$ is arbitrary, the claim follows. □

The four missing operations:

$$+\infty - \infty, \quad 0 \cdot \infty, \quad \frac{0}{0} \quad \text{and} \quad \frac{\infty}{\infty}$$

are called **indeterminate forms**, since no rule can be established a priori to determine their result.

Example 5: Resolving indeterminate forms $+\infty - \infty$

Consider the sequence

$$n \mapsto \sqrt{n+1} - \sqrt{n-1}$$

We have the difference of two sequences:

$$n \mapsto \sqrt{n+1} \quad \text{with} \quad \lim_{n \rightarrow +\infty} \sqrt{n+1} = +\infty$$

$$n \mapsto \sqrt{n-1} \quad \text{with} \quad \lim_{n \rightarrow +\infty} \sqrt{n-1} = +\infty$$

Hence we fall into the indeterminate form $+\infty - \infty$.

Multiplying and dividing by $\sqrt{n+1} + \sqrt{n-1}$ we obtain

$$\frac{(\sqrt{n+1} - \sqrt{n-1})(\sqrt{n+1} + \sqrt{n-1})}{\sqrt{n+1} + \sqrt{n-1}} = \frac{(\sqrt{n+1})^2 - (\sqrt{n-1})^2}{\sqrt{n+1} + \sqrt{n-1}} = \frac{2}{\sqrt{n+1} + \sqrt{n-1}}$$

recalling that $a^2 - b^2 = (a - b)(a + b)$. Now, considering the sequence in the denominator, we have

$$\lim_{n \rightarrow +\infty} \sqrt{n+1} + \sqrt{n-1} = +\infty + \infty = +\infty$$

Using the rule

$$a_n \rightarrow \ell_a \in \mathbb{R}, \quad c_n \rightarrow \infty \Rightarrow \frac{a_n}{c_n} \rightarrow 0$$

of the theorem on the partial arithmetization of the infinity symbol (product), we have:

$$\lim_{n \rightarrow +\infty} \sqrt{n+1} - \sqrt{n-1} = \lim_{n \rightarrow +\infty} \frac{2}{\sqrt{n+1} + \sqrt{n-1}} = \frac{2}{+\infty} = 0$$

- Limits of sequences of the form:

$$\{a_n^{b_n}\}$$

can be handled by considering the sequence of their logarithms, taking for simplicity the base e .

Given a sequence $\{a_n^{b_n}\}$, we have that:

$$\text{if } b_n \log a_n \rightarrow \ell \text{ then } a_n^{b_n} \rightarrow e^\ell \quad (3)$$

$$\text{if } b_n \log a_n \rightarrow +\infty \text{ then } a_n^{b_n} \rightarrow +\infty \quad (4)$$

$$\text{if } b_n \log a_n \rightarrow 0 \text{ then } a_n^{b_n} \rightarrow 1 \quad (5)$$

$$\text{if } b_n \log a_n \rightarrow -\infty \text{ then } a_n^{b_n} \rightarrow 0 \quad (6)$$

If the sequence $\{b_n \log a_n\}$ is indeterminate (has no limit), then $\{a_n^{b_n}\}$ is indeterminate as well.

Example 6: Computing limits by taking logarithms

$$\lim_{n \rightarrow +\infty} (3n)^{(-3n^2+7)}$$

taking logarithms we have

$$\underbrace{(-3n^2+7)}_{b_n} \log \underbrace{3n}_{a_n}$$

since

$$\lim_{n \rightarrow +\infty} (-3n^2+7) \log(3n) = -\infty \cdot +\infty = -\infty$$

then

$$\lim_{n \rightarrow +\infty} (3n)^{(-3n^2+7)} = 0$$

Example 7: Computing limits by taking logarithms (alternative method)

$$\lim_{n \rightarrow +\infty} (3n)^{(-3n^2+7)} = \lim_{n \rightarrow +\infty} e^{\log((3n)^{(-3n^2+7)})} = \lim_{n \rightarrow +\infty} e^{(-3n^2+7) \log(3n)}$$

since

$$\lim_{n \rightarrow +\infty} (-3n^2+7) \log(3n) = -\infty \cdot +\infty = -\infty$$

then

$$\lim_{n \rightarrow +\infty} (3n)^{(-3n^2+7)} = e^{-\infty} = 0$$

We also have the following **indeterminate forms**:

$$1^\infty, \quad 0^0, \quad \infty^0$$

Taking logarithms, they correspond to the indeterminate form

$$0 \cdot \infty$$

since:

$$\log(1^\infty) = \infty \log 1 = \infty \cdot 0$$

$$\log(0^0) = 0 \log 0 = 0 \cdot -\infty$$

$$\log(+\infty^0) = 0 \log(+\infty) = 0 \cdot +\infty$$

Finally, since $-\infty^0 = -1 \cdot (+\infty^0)$, we have

$$-1 \cdot (\log(+\infty^0)) = -1 \cdot (0 \log +\infty) = -1 \cdot (0 \cdot +\infty)$$

Limits of the form

$$0^{+\infty} \text{ and } 0^{-\infty}$$

are not indeterminate forms. We have:

$$0^{+\infty} = 0 \text{ and } 0^{-\infty} = +\infty.$$

since, taking logarithms, we have

$$\log(0^{+\infty}) = +\infty \cdot \log 0 = +\infty \cdot -\infty = -\infty \quad \text{and} \quad e^{-\infty} = 0$$

$$\log(0^{-\infty}) = -\infty \cdot \log 0 = -\infty \cdot -\infty = +\infty \quad \text{and} \quad e^{+\infty} = +\infty$$

Example 8: Limits of the forms $0^{+\infty}$ and $0^{-\infty}$

- Example of the case $0^{+\infty} = 0$:

$$\lim_{n \rightarrow +\infty} \left(\frac{1}{n}\right)^{\log n} = 0^{+\infty} = 0.$$

Indeed we have

$$\left(\frac{1}{n}\right)^{\log n} = e^{\log\left(\frac{1}{n}\right)^{\log n}} = e^{\log n \log \frac{1}{n}}$$

hence

$$\lim_{n \rightarrow +\infty} \left(\frac{1}{n}\right)^{\log n} = \lim_{n \rightarrow +\infty} e^{\overbrace{\log n}^{\rightarrow +\infty} \overbrace{\log \frac{1}{n}}^{\rightarrow -\infty}} = e^{-\infty} = 0.$$

- Example of the case $0^{-\infty} = +\infty$:

$$\lim_{n \rightarrow +\infty} \left(\frac{1}{n}\right)^{-\log n} = 0^{-\infty} = +\infty.$$

Indeed we have

$$\left(\frac{1}{n}\right)^{-\log n} = e^{\log\left(\frac{1}{n}\right)^{-\log n}} = e^{-\log n \log \frac{1}{n}}$$

hence

$$\lim_{n \rightarrow +\infty} \left(\frac{1}{n}\right)^{-\log n} = \lim_{n \rightarrow +\infty} e^{\overbrace{-\log n}^{\rightarrow -\infty} \overbrace{\log \frac{1}{n}}^{\rightarrow -\infty}} = e^{+\infty} = +\infty.$$